

Root systems

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0.1 Root pairings

0.1.1 Definitions

Recall that a *perfect pairing* is a quadruple $(R, M, N, \mathcal{L})$ where R is a ring, M and N are R -modules, and $\mathcal{L} : M \times N \rightarrow R$ is a bilinear map such that the induced maps $M \rightarrow \text{Hom}_R(N, R)$ and $N \rightarrow \text{Hom}_R(M, R)$ are isomorphisms of R -modules.

If I is a set, denote by $\text{Perm}(I)$ the group of permutations of I .

Definition 1. A *root pairing* is a tuple $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ where $(R, M, N, \mathcal{L})$ is a perfect pairing, I is a set, $\alpha : I \rightarrow M, i \mapsto \alpha_i$, $\alpha^\vee : I \rightarrow N, i \mapsto \alpha_i^\vee$, and $s : I \rightarrow \text{Perm}(I), i \mapsto s_i$, with the conditions that

1. $\alpha, \alpha^\vee$ are injective,
2. For all $i \in I$, $\mathcal{L}(\alpha_i, \alpha_i^\vee) = 2$,
3. For all $i, j \in I$, we have

$$\begin{aligned}\alpha_{s_i(j)} &= \alpha_j - \mathcal{L}(\alpha_j, \alpha_i^\vee)\alpha_i, \\ \alpha_{s_i(j)}^\vee &= \alpha_j^\vee - \mathcal{L}(\alpha_i, \alpha_j^\vee)\alpha_i^\vee.\end{aligned}$$

0.1.2 Root pairing of type A

Consider the perfect pairing $(R, M, N, \mathcal{L}) = (\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, (\cdot, \cdot))$ where $(\cdot, \cdot)$ is the canonical pairing between $\mathbb{Z}^n$ and its algebraic dual $(\mathbb{Z}^n)^* := \text{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{Z})$, i.e. the dot product.

We denote by $e_1, \dots, e_n$ the standard basis of $\mathbb{Z}^n$ and by $e_1^*, \dots, e_n^*$ the dual basis of $(\mathbb{Z}^n)^*$.

Write $[1, n] = \{1, \dots, n\}$. An *interval* in $[1, n]$ is a subset of the form $[i, j] \stackrel{\text{denotes}}{=} \{i, i+1, \dots, j\}$ for some $1 \leq i \leq j \leq n$. We denote by I_n the set of intervals in $[1, n]$. Write $[i] := [i, i]$ for $1 \leq i \leq n$.

A *signed interval* in $[1, n]$ is an element of $I_n := I_n \times \{1, -1\}$. We denote by I_n^+ the set of signed intervals of the form $(J, 1)$ and by I_n^- the set of signed intervals of the form $(J, -1)$. It is clear that $I_n = I_n^+ \sqcup I_n^-$. Throughout the rest of this file, we write J for the signed interval $(J, 1)$ and $-J$ for $(J, -1)$.

Let $A = (a_{i,j})$ be the Cartan matrix of type A_n defined by $a_{i,i} = 2$ for all $i \in [1, n]$, $a_{i,j} = -1$ if $|i - j| = 1$, and $a_{i,j} = 0$ otherwise.

Define

$$\begin{aligned}\alpha : I_n &\rightarrow \mathbb{Z}^n, & \forall J \in I_n^+, \quad \alpha_{\pm J} &= \pm \left(\sum_{j \in J} A e_j \right), \\ \alpha^\vee : I_n &\rightarrow (\mathbb{Z}^n)^*, & \forall J \in I_n^+, \quad \alpha_{\pm J}^\vee &= \pm \left(\sum_{j \in J} e_j^* \right).\end{aligned}$$

Finally, define $s : I_n^+ \rightarrow \text{Perm}(I_n)$ as follows. Fix $J, K \in I_n^+$, and set $L = (J \cup K) \setminus (J \cap K)$. Then J, K, L satisfy exactly one of the following four conditions.

R1. $J = K$.

R2. $J \subset K$ and $L \in I_n$.

R3. $K \subset J$ and $L \in I_n$.

R4. $L \notin I_n$.

Then,

$$s_J(K) = \begin{cases} -K & \text{in Case } \mathbf{R1}, \\ K & \text{in Case } \mathbf{R2}, \\ L & \text{in Case } \mathbf{R3}, \\ -L & \text{in Case } \mathbf{R4}. \end{cases} \quad (1)$$

We extend s_J to I_n by setting $s_J(-K) = -s_J(K)$ for all $K \in I_n^+$. We extend s to I_n by setting $s_{-J} = s_J$ for all $J \in I_n^+$.

Lemma 2. *For all J , we have $s_J \in \text{Perm}(I_n)$.*

Proof. Fix $J \in I_n$. We prove that $s_J^2 = \text{id}$ by a straightforward case-by-case check using the definition of s_J and the four cases above. The claim then follows by noting that $s_J^{-1} = s_J$. $\square$

Theorem 3. *The 8-tuple $(\mathbb{Z}, \mathbb{Z}^l, (\mathbb{Z}^l)^*, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ is a root pairing of A_l , where:*

1. $\mathcal{L}$ is the dot product on $\mathbb{Z}^l \times (\mathbb{Z}^l)^*$
2. $I = \{\text{signed intervals on the set } \{1, \dots, n\}\} = I^+ \cup I^-$
3. $s : I \rightarrow \text{Perm}(I)$ where for $J \in I$, $s_J = s_{-J}$, $s_J(-K) = -s_J(K)$, defined as $s_J(K) =$

$$\begin{cases} J \cup K & \text{if } J \cap K = \emptyset \wedge J \cup K \in I \\ K & \text{if } J \cap K = \emptyset \wedge J \cup K \notin I \\ K \setminus J & \text{if } J \cap K \neq \emptyset \wedge J \subset K \wedge K \setminus J \in I \\ K & \text{if } J \cap K \neq \emptyset \wedge J \subset K \wedge K \setminus J \notin I \\ -K & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K = \emptyset \\ -(J \setminus K) & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K \in I \\ K & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K \notin I \\ K & \text{if } J \cap K \neq \emptyset \wedge K \not\subseteq J \wedge J \not\subseteq K \end{cases}$$

Equivalently (for use in Lean),

1. $I = \{(i, j, \varepsilon) | 1 \leq i \leq j \leq l, \varepsilon = \pm 1\}$
2. $\alpha : I \hookrightarrow \mathbb{Z}^l, (i, j, \varepsilon) \mapsto \varepsilon \left(\sum_{k=i}^j A_l e_k \right)$ (where A_l is the Cartan Matrix of A_l)
3. $\alpha^\vee : I \hookrightarrow (\mathbb{Z}^l)^*, (i, j, \varepsilon) \mapsto \varepsilon \left(\sum_{k=i}^j e_k^\top \right)$
4. $s : I \rightarrow \text{Perm}(I)$ where $s(i, j, 1) = s(i, j, -1)$, $s(i, j, \varepsilon)(k, l, 1) = (p, q, \eta)$ means that $s(i, j, \varepsilon)(k, l, -1) = (p, q, -\eta)$, and

$$s(i, j, 1)(k, l, 1) = \begin{cases} (k, l, -1) & \text{if } (i, j) = (k, l) \\ (l+1, j, -1) & \text{if } i = k, j > l \\ (j+1, l, 1) & \text{if } i = k, j < l \\ (i, k-1, -1) & \text{if } i < k, j = l \\ (k, i-1, 1) & \text{if } i > k, j = l \\ (k, j, 1) & \text{if } i = l+1 \\ (i, l, 1) & \text{if } k = j+1 \\ (k, l, 1) & \text{otherwise} \end{cases}$$

Proof.

□